Chinmay Arora

Computational biology · Cell therapy · Pharmaceutical sciences

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PROFILE

Biomedical engineer and registered pharmacist (B.Pharm, DIPSAR; M.Tech, IIT Guwahati, CPI 9.11/10). M.Tech research developed an end-to-end in silico pipeline for dual-target CAR-NK cell design, integrating TCGA transcriptomics, DESeq2 differential expression, surfaceome and normal-tissue safety filtering, AlphaFold2/3 structure prediction and HADDOCK docking. Experience spans molecular biology research at AIIMS New Delhi, pharmacovigilance and pharmacogenomics data curation, RP-UFLC analytical method development, bulk RNA-seq workflows, quantum machine learning and medical-device prototyping, complemented by teaching and student leadership. GATE Life Sciences AIR 696.

EDUCATION

M.Tech, Biomedical Science and Engineering

Jul 2024 – Jul 2026

Indian Institute of Technology Guwahati — Jyoti and Bhupat Mehta School of Health Sciences and Technology · CPI 9.11 / 10

- Specialisation: Regenerative Medicines, Stem Cells and Therapeutics.
- Thesis: *Design and Optimization of Dual CAR-NK Cells for Breast Cancer Therapy Using AI/ML Approaches* (Supervisors: Prof. Bithiah Grace Jaganathan and Dr. Suresh Sundaram).

B.Pharm

Dec 2020 – Jun 2024

Delhi Institute of Pharmaceutical Sciences and Research (DIPSAR), DPSRU, New Delhi · CGPA 7.80 / 10

RESEARCH EXPERIENCE

M.Tech thesis — dual CAR-NK design for breast cancer

Jul 2025 – Jun 2026

Stem Cell and Cancer Biology Group, IIT Guwahati

- Built a two-phase in silico pipeline to identify, structurally validate and rank dual-antigen targets for tandem CAR-NK (tanCAR-NK) cells across PAM50 breast cancer subtypes.
- Acquired TCGA-BRCA STAR counts from the GDC (1,224 samples: 1,111 primary tumour, 113 solid-tissue normal), filtered to 16,834 protein-coding genes and assigned PAM50 subtypes with genefu.
- Ran subtype-versus-normal DESeq2 analysis (padj < 0.05, log2FC ≥ 1) giving 2,186–3,180 upregulated genes per subtype, with complementary limma-voom and GEO2R analysis.
- Intersected DEGs with a 2,304-gene HPA surfaceome list, applied a four-tier GTEx v8 normal-tissue safety classification (325 Tier A antigens) and CARTAR logic-gate pairing; literature and patent tractability screening yielded 9 final tanCAR pairs.
- Modelled scFvs for 7 antigens in both VH–(G4S)3–VL and VL–(G4S)3–VH orientations with AlphaFold2 (ColabFold, mean pLDDT > 90), annotating CDRs with SAbPred/IMGT and membrane topology with DeepTMHMM.
- Validated antigen–scFv interfaces through CDR-restrained HADDOCK 2.4 docking (ClusPro as a blind check) and AlphaFold3 three-chain co-docking for 6 pairs; designed a six-parameter weighted composite score ranking CEACAM5–CLDN9 and PSCA–CEACAM5 as the top Grade A candidates.
- Generated de novo KISS1R binder scaffolds with RFdiffusion and ProteinMPNN, and assembled complete NK-optimised CAR sequences (CD8α signal peptide and hinge/TM, 2B4 co-stimulatory domain, CD3ζ) for 4 constructs.

Research Intern, Department of Preventive Oncology

May – Jul 2023

Dr. B.R.A. Institute Rotary Cancer Hospital, AIIMS New Delhi · Supervisor: Dr. Pawan Kumar

- Secured the only intern position nationwide under the Scientific Social Responsibility scheme of DST-SERB, supported by a ₹10,000 research grant.
- Contributed to a project developing a cancer immunotherapeutic that incorporates the cancer/testis antigen NY-ESO-1 into *Mycobacterium indicus pranii*.
- Performed PCR, agarose gel electrophoresis, RNA and plasmid isolation, bacterial culture, transformation, electroporation and SDS-PAGE, supporting 10+ experiments.

Genetic underpinnings of adverse events across populations

2023 – 2025

Department of Pharmacology, DIPSAR/DPSRU · Guide: Dr. Anoop Kumar · Co-author with Asees Singh

- Co-conceptualised a population-level framework testing whether the adverse drug reaction profiles of four ACE inhibitors are associated with genetic variation across the US, UK and Canada.
- Curated the analysis datasets: 3,000+ OpenVigil (FDA FAERS) adverse-event records stratified by region, sex and age group, SmPC-listed ADRs, MedDRA organ-system classes and 1000 Genomes variant data.
- Post-processed disproportionality and correlation analyses and refined the manuscript draft; the study found a single significant ADR–gene correlation after adjustment, indicating no population-level genetic association. Manuscript awaiting review by the corresponding author.

B.Pharm project — RP-UFLC method for caffeine quantification

Mar – Jun 2024

Department of Pharmaceutics, DIPSAR/DPSRU · Guide: Dr. Beauty Behera

- Developed a reversed-phase UFLC method for caffeine in a commercial soft drink on a Hyperclone 5 µm BDS C18 column with acetonitrile : phosphate buffer (11:89 v/v) at 0.5 mL/min.
- Evaluated ICH Q2(R1) validation parameters: linearity over 20–100 ppm, LOD and LOQ, intra- and inter-day precision and spike recovery.

TEACHING

Teaching Assistant

Jul 2025 – Aug 2026

Indian Institute of Technology Guwahati

- TA for HT105 Introduction to Computing, HT506 Biotechniques and Bioinstrumentation Laboratory and DAO3023 Applied Machine Learning for Signal Processing (B.Sc. (Hons) DS&AI online programme).
- Designed class material on Python lists and conditional statements and taught Python and R to 90+ students.
- Ran hands-on bioinformatics tutorials, including R programming and survival analysis, for Master's and PhD students.

TECHNICAL PROJECTS

Arogyavitals — maternal health monitoring prototype

Jan – Apr 2025

Product Design & Prototyping Laboratory, JBMSHST, IIT Guwahati · Team Clinovators (4 members)

- Co-developed a portable Arduino Uno R4 WiFi prototype measuring blood pressure, heart rate and body temperature, with solar, battery and USB-C power for off-grid rural use.
- Led software development: MitraLink, a Flutter (Dart) Android app that logs vitals offline, exports records to Excel and links users to the eSanjeevani telemedicine portal.
- Estimated a bill of materials of ₹5,528, over 40% below the reported cost of a comparable published device (Dese et al., HardwareX, 2022). Prototype not clinically validated.

Bulk RNA-seq analysis pipeline

Sep – Oct 2025

Self-directed · GEO GSE106305 (LNCaP and PC-3 prostate cancer cells, hypoxia vs normoxia)

- Processed ~60 GB of raw sequencing data (20 SRA runs into 8 samples) through a documented, reproducible command-line workflow on WSL/Ubuntu.
- Chained SRA Toolkit, FastQC/MultiQC, Trimmomatic, HISAT2 (GRCh38), SAMtools, Qualimap and featureCounts, with Python scripts automating downloads and count-matrix assembly.
- Ran DESeq2 in R (VST, PCA, clustering, heatmaps) on a normalised matrix of 17,462 genes and ranked the top 20 enriched pathways with ReactomePA.

Collision risk prediction for space debris using quantum machine learning Jun – Jul 2025

Quantum Tech Community and 4i Labs, IIT Guwahati — course final assignment

- Built a quantum support vector classifier in Qiskit (ZZFeatureMap kernel, Aer simulator) on 960 training samples and 12 features, benchmarked against a classical RBF-kernel SVM.
- Completed an 8-week course on quantum computing techniques and algorithms; awarded a Certificate of Excellence.

TECHNICAL SKILLS

Transcriptomics: SRA Toolkit, FastQC, MultiQC, Trimmomatic, HISAT2, SAMtools, Qualimap, featureCounts, DESeq2, edgeR, limma-voom, Bioconductor, genefu, ReactomePA/GSEA, network and enrichment analysis.

Structural biology and protein design: AlphaFold2 (ColabFold), AlphaFold3, HADDOCK, ClusPro, RFdiffusion, ProteinMPNN, scFv design, CDR annotation (SAbPred/IMGT), DeepTMHMM, PyMOL, AutoDock4/AutoDockTools, Open Babel, LigPlot+.

Databases: TCGA/GDC, GEO-NCBI (GEO2R), GTEx, Human Protein Atlas, MSigDB, RCSB PDB, KEGG, 1000 Genomes, OpenVigil/FDA FAERS.

Programming and data: Python (NumPy, pandas, Matplotlib, scikit-learn, Qiskit), R (ggplot2, pheatmap), SQL/MySQL, Bash, Dart/Flutter, C++ (basic), HTML, Git, OpenRefine.

Molecular biology (hands-on): PCR, agarose gel electrophoresis, RNA and plasmid isolation, bacterial culture, transformation, electroporation, SDS-PAGE; mammalian cell culture, flow cytometry, Western blot, molecular cloning and 3-D bioprinting from M.Tech training.

Pharmaceutical sciences: RP-UFLC/HPLC method development, ICH Q2(R1) validation, pharmacovigilance with MedDRA classification, design of experiments and QbD (JMP).

Design and media: Figma, Adobe Photoshop, Illustrator, Lightroom, Premiere Pro, Affinity, Canva; photography.

LICENCES, AWARDS AND CERTIFICATIONS

- **Registered Pharmacist**, Punjab Pharmacy Council, Government of Punjab (valid until Dec 2029).
- **GATE Life Sciences (XL)** — All India Rank 696.
- **First runner-up, MedTechNova'25** Medical Device Development Hackathon, SIBDS-IIBC and NewGen IEDC with AIIMS Guwahati at the Research & Industrial Conclave 2025 (Oct 2025).
- **Certificate of Excellence**, Quantum Machine Learning Course 2025, QT IITG and 4i Labs.
- Global Consumer Intelligence Course 2025, Matsuo-Iwasawa Lab, The University of Tokyo (Jan 2026); General Course on Intellectual Property, WIPO Academy (Nov 2024).
- Third place, poster competition on antibiotic use and resistance, DPSRU (May 2023).

WRITTEN OUTPUTS

- Arora C. *Design and Optimization of Dual CAR-NK Cells for Breast Cancer Therapy Using AI/ML Approaches.* M.Tech thesis, IIT Guwahati, 2026.
- Kumar A, Singh A, Arora C. *Genetic Underpinnings of Adverse Events Across Populations: The Case of Anti-Hypertensive Drugs.* Manuscript in preparation.
- Arora C. *Molecular Docking Using AutoDock4* — laboratory protocol for the Stem Cell and Cancer Biology Group, IIT Guwahati.
- Mukherjee R, Arora C, Basu SK, Roychowdhury P. *Arogyavitals: Smart, Affordable Maternal Health Monitoring for Rural India.* Project poster, JBMSHST, IIT Guwahati, 2025.

LEADERSHIP AND RESPONSIBILITY

- **Content & Media Head**, Research & Industrial Conclave 2025, IIT Guwahati (Aug – Oct 2025).
- **Peer Mentor**, Saathi Counselling Cell, IIT Guwahati (Sep 2025 – Jul 2026).
- **North Divisional Head**, Indian Pharmaceutical Association Students' Forum, National Executive Committee (Jan 2024 – Apr 2025).
- **Vice-Chairperson**, IPA-SF Delhi State Chapter (May – Dec 2023) — re-established the chapter and served on its first council.

- **Vice-President**, Paridrishya, the photography society of DIPSAR/DPSRU (Nov 2022 – Oct 2023).
- **Media & Publicity Chair**, Rotaract Club of DPSRU (Jul 2022 – Jun 2023).

LANGUAGES

Punjabi (native) · Hindi (full professional) · English (professional working) · Japanese (elementary)